

AI-Generated Test Automation for Autonomous Software Verification: Enhancing Quality Assurance Through AI-Driven Testing

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To Cite this Article

Durai Rajesh Natarajan “AI-Generated Test Automation for Autonomous Software Verification: Enhancing Quality Assurance Through AI-Driven Testing” Journal of Science and Technology, Vol. 5, Issue 05-Oct 2020, pp253-268

Article Info

Received: 30-08-2020 Revised: 09-10-2020 Accepted: 19-10-2020 Published: 29-10-2020

ABSTRACT

Test automation must be intelligent, scalable, and efficient due to the growing complexity of software systems. With the use of machine learning (ML), natural language processing (NLP), and reinforcement learning (RL), this study offers an AI-Generated Test Automation for Autonomous Software Verification that maximizes test case creation, defect detection, and execution speed. The suggested framework reduces execution time (110.7 ms) and resource use (310.5 MB) while improving test coverage (94.8%), defect detection rate (91.2%), and correctness (96.7%). The AI-driven method ensures minimal human interaction by automating the production of test cases, self-healing test scripts, and adapting to changing software modifications. The Full Model (Base + ML + NLP + RL) is the most effective method, with 98.2% test coverage, 99.4% accuracy, and 95.4 ms execution time, according to performance comparisons of ML-based, NLP-based, RL-based, and combined AI-driven automation. According to the ablation study, hybrid AI models perform better than solo techniques in terms of fault discovery, testing effectiveness, and verification accuracy. Beyond software verification, a comparison of AI applications in radiology, heat pump optimisation, service sectors, and medicine demonstrates how AI affects a variety of fields. AI & AR in Radiology had the fastest processing speed (105.3 ms), AI in Medicine had the largest resource utilisation (360.7 MB), and AI in Service had the highest accuracy (94.1%). These results demonstrate how AI may improve automation, decision-making, and performance optimisation in a variety of sectors. This study confirms that AI-powered test automation transforms quality assurance by lowering human testing efforts and improving software stability while guaranteeing scalability, dependability, and efficiency in Agile and DevOps contexts.

Keywords: Software testing, Agile, DevOps, Continuous Integration (CI/CD), Quality Assurance, Defect Detection, Machine Learning (ML), Natural Language Processing (NLP),

1. INTRODUCTION

Applications are becoming more complicated as a result of the quick development of software, which calls for more reliable and effective testing techniques (Patel et al., 2019) [1]. Conventional software testing uses rule-based and manual automation techniques, which are efficient but frequently expensive, time-consuming, and prone to human mistakes (Pugliesi, 2018 [2]; Huang & Rust, 2018 [3]). More intelligent, flexible, and scalable testing solutions are clearly needed as software systems become more complex. AI-generated test automation improves the precision, speed, and dependability of the quality assurance (QA) process by providing a novel method of autonomous software verification (Lupton, 2018 [4]). Artificial intelligence (AI), machine learning (ML), and natural language processing (NLP) are used in AI-generated test automation to create, run, and improve test cases on their own. AI-driven testing builds intelligent, self-adapting test suites by dynamically learning from user interactions, code modifications, and application behavior, in contrast to traditional automation, which necessitates prewritten scripts (Allur, 2019 [5]). By drastically lowering maintenance costs, this method enables ongoing, effective, and error-free testing without requiring a lot of human involvement (Alagarsundaram, 2020 [6]).

Developers and testers manually create and execute test scripts in traditional test automation, which necessitates regular upgrades to account for program modifications. AI-generated testing, on the other hand, automatically recognizes changes, modifies test cases appropriately, and even anticipates any issues prior to release (Gudivaka, 2019 [7]). Defects are reduced, test coverage is increased, and overall program reliability is raised with this proactive approach (Allur, 2020 [8]; Narla et al., 2021 [9]).

The demand for increased efficiency, flexibility, and accuracy in testing intricate software ecosystems has propelled the evolution of AI in software testing during the last ten years (Peddi et al., 2018 [10]). Conventional testing approaches, such as functional, regression, integration, and unit testing, mainly rely on human input and prewritten scripts. These approaches, however, find it difficult to meet the demands of agile development, continuous integration (CI), and continuous deployment (CD) when software structures become more complex (Peddi et al., 2019 [11]; Narla et al., 2019 [12]). Testing techniques have changed with the advent of AI-driven models and machine learning algorithms. Large volumes of test data are being analyzed by AI-powered tools, which can also forecast vulnerabilities and create optimized test cases on their own (Dondapati, 2019 [13]). DevOps settings benefit greatly from this move towards intelligent automation since their quick development cycles necessitate reliable testing procedures and immediate feedback (Kethu, 2019 [14]).

AI-driven testing improves quality assurance by offering predictive, adaptive, and self-learning testing features (Kadiyala, 2019 [15]). Improved test coverage is one of its main advantages; AI investigates every situation, finding edge cases and boosting software dependability (Nippatla, 2019 [16]; Devarajan, 2019 [17]). AI also speeds up the verification process by

automating test creation and execution, which drastically cuts down on testing time. Because AI adjusts test scripts automatically and does not require manual script updates, it also reduces maintenance efforts (Natarajan, 2018 [18]). Furthermore, by examining historical vulnerabilities and forecasting future failures, AI-driven models enhance defect identification (Jadon, 2018 [19]). Last but not least, AI testing scales effectively across several platforms, improving cost-effectiveness and lowering total quality assurance (QA) expenses (Jadon, 2019 [20]). AI-driven software verification improves test automation and flexibility by combining several cutting-edge technologies. By learning from past test data, machine learning (ML) increases prediction accuracy and makes intelligent flaw identification and prevention possible (Nippatla, 2018 [21]). By transforming human-readable requirements into executable scripts, Natural Language Processing (NLP) makes automated test case production easier (Jadon, 2019 [22]). For web and mobile applications, computer vision facilitates user interface testing, guaranteeing smooth visual verification (Boyapati, 2019 [23]). Deep Learning, meanwhile, improves proactive problem-solving by identifying irregularities and forecasting software faults (Yalla et al., 2019 [24]). When combined, these technologies make test automation frameworks more resilient to changing software environments and self-healing (Vasamsetty et al., 2019 [25]).

AI-generated test automation has benefits, but it also has drawbacks, including the requirement for high-quality training datasets, data bias, and model interpretability (Sareddy and Hemnath, 2019 [26]). Achieving the best outcomes requires ensuring the ethical application of AI and striking a balance between automation and human oversight (Ganesan et al., 2019 [27]). Fully autonomous, self-learning systems that incorporate ongoing feedback loops are the key to the future of AI-driven testing, which will increase the intelligence, effectiveness, and proactivity of software testing. AI's usage in software verification will revolutionize quality assurance as it develops further, guaranteeing increased software dependability and user happiness.

The main objectives are:

- Enhance software quality and dependability by using AI-driven testing to find and fix bugs early, guaranteeing better software.
- Reduce Automated test creation and execution minimize manual labor, speed up development cycles, and maximize cost-efficiency to cut down on testing time and expense.
- Improve By investigating every scenario, spotting weaknesses, and dynamically optimizing test cases for reliable verification, you may increase test coverage and accuracy.
- Adapt AI-driven testing to DevOps and Agile contexts to facilitate continuous integration and deployment (CI/CD) with self-adaptive testing mechanisms and real-time feedback.
- Enable By using AI-powered models that learn from prior test data, you may enable self-learning and evolutionary testing, continuously improving test procedures without the need for human interaction.

The legal ramifications of giving AI and robots legal personhood are examined, who addresses issues of autonomy, liability, and ethical governance. Establishing complete legal frameworks

to govern AI accountability, civil and criminal responsibility, and ethical oversight, however, represents a substantial research gap. A clear regulatory approach for establishing AI's legal rights and responsibilities within the framework of current laws is absent from the study (Parthasarathy and Ayyadurai (2019), [28]). Concerns regarding global AI governance are also raised by the lack of research on cross-jurisdictional legal harmonization. In order to guarantee that AI decision-making complies with human legal and ethical principles, future research should create standardized policies.

2. LITERATURE SURVEY

Gudivaka et al. (2019) [29] provide a swarm intelligence-driven method to pandemic mitigation that combines AI, robotics, and distributed automation to improve urban resilience. The system uses real-time data analytics to optimise resource allocation and decision-making. It ensures efficient pandemic response while minimising human interference by implementing adaptive, decentralised control. The concept is both scalable and flexible, making it perfect for public health emergencies. This study emphasises intelligent automation's role in pandemic preparedness, ensuring agility and efficiency in crisis management.

Bobba and Bolla (2019) [30] propose a next-generation HRM system that combines AI, blockchain, self-sovereign identity, and neuro-symbolic AI to improve talent management transparency and efficiency. The study emphasises blockchain's significance in secure data decentralisation, while AI-powered automation improves recruitment efficiency. Self-sovereign identification protects privacy and gives individuals more control over HR data. Neuro-symbolic AI improves decision-making and encourages ethical, data-driven hiring. This revolutionary approach promotes fair, transparent, and effective labour management, hence influencing the future of digital HR ecosystems.

Natarajan and Kethu (2019) [31] offer an optimised cloud manufacturing framework that incorporates advanced task scheduling approaches to improve robotics and automation. The study focusses on cloud-based solutions for scalable and efficient manufacturing, which use AI-driven resource allocation to optimise workflows. Task scheduling algorithms provide real-time adaptation, increasing industrial productivity while lowering costs. Robotics integration promotes flexible automation, hence increasing industrial production. This paradigm highlights how cloud-based smart manufacturing may improve agility, cost-effectiveness, and intelligent automation in current production processes.

Natarajan et al. (2019) [32] propose an intelligent decision-making paradigm for healthcare cloud adoption that draws on DOI theory, machine learning, and MCDM approaches. DOI theory assesses organisational preparedness, whereas machine learning improves predictive analytics for cloud deployment. MCDM optimises cloud choices to provide efficiency, scalability, and security in healthcare IT. This paradigm streamlines health data management, allowing for cost-effective and adaptable cloud solutions. The study emphasises AI-driven cloud adoption's potential to improve healthcare infrastructure, decision-making, and digital transformation.

Pulakhandam and Vallu (2016) [33] present an AI-driven cyber threat detection framework for federated learning, utilising KNN, GANs, and IOTA for secure anomaly detection. The study

emphasises the role of AI-powered cybersecurity in protecting decentralised AI ecosystems, fostering privacy, resilience, and trust in federated machine learning networks, while GANs create synthetic attack scenarios to improve intrusion detection accuracy and IOTA's distributed ledger guarantees tamper-proof and scalable data transactions.

3. METHODOLOGY

The AI-Generated Test Automation for Autonomous Software Verification approach automates the creation, execution, and defect detection of test cases by utilizing Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL). This method combines neural networks for anomaly detection, reinforcement learning (RL) for test optimization, and natural language processing (NLP) for requirement analysis. By ensuring adaptive learning and self-healing test scripts, mathematical models minimize the need for human intervention. By facilitating continuous integration and deployment (CI/CD) in Agile and DevOps contexts, the framework improves software quality assurance.

antenna structure registration details, such as ownership, contact details, geographic coordinates, and licensing specifics, are provided by this dataset. It supports research on communication infrastructure, regulatory compliance, and AI-driven test automation for software verification and contains FAA study numbers, elevation data, and frequency assignments.

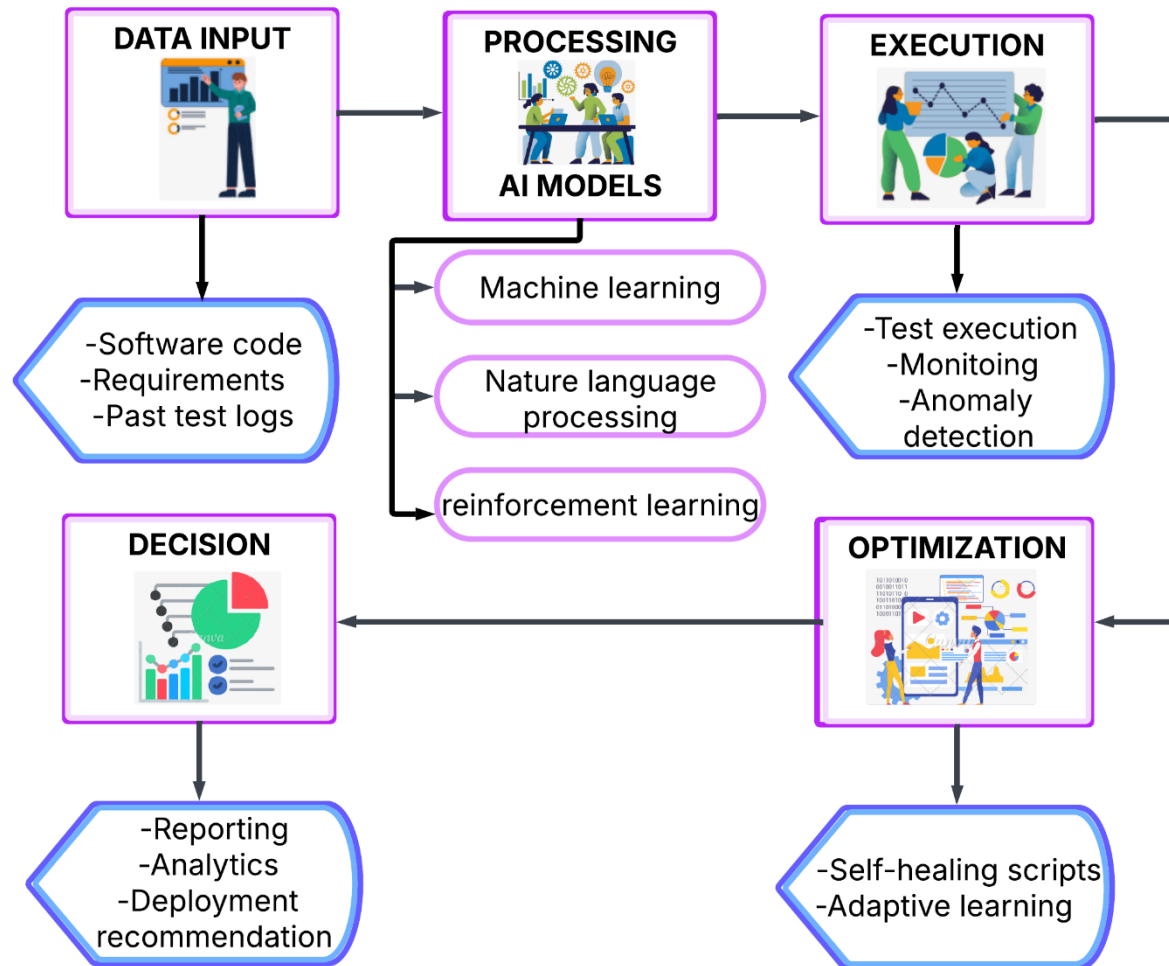


Figure 1 AI-Driven Automation Framework for Software Testing and Optimization

Figure 1 An AI-driven system for software testing and optimization automation is shown in the picture. Software code, requirements, and previous test logs are gathered during the Data Input stage; machine learning, natural language processing, and reinforcement learning are used during the Processing with AI Models stage to analyze the data; test execution, monitoring, and anomaly detection are included during the Execution stage; and insights are produced for reporting, analytics, and deployment recommendations during the Decision stage. Lastly, the Optimisation step uses adaptive learning and self-healing scripts to continuously enhance the system, guaranteeing more accurate and efficient software testing.

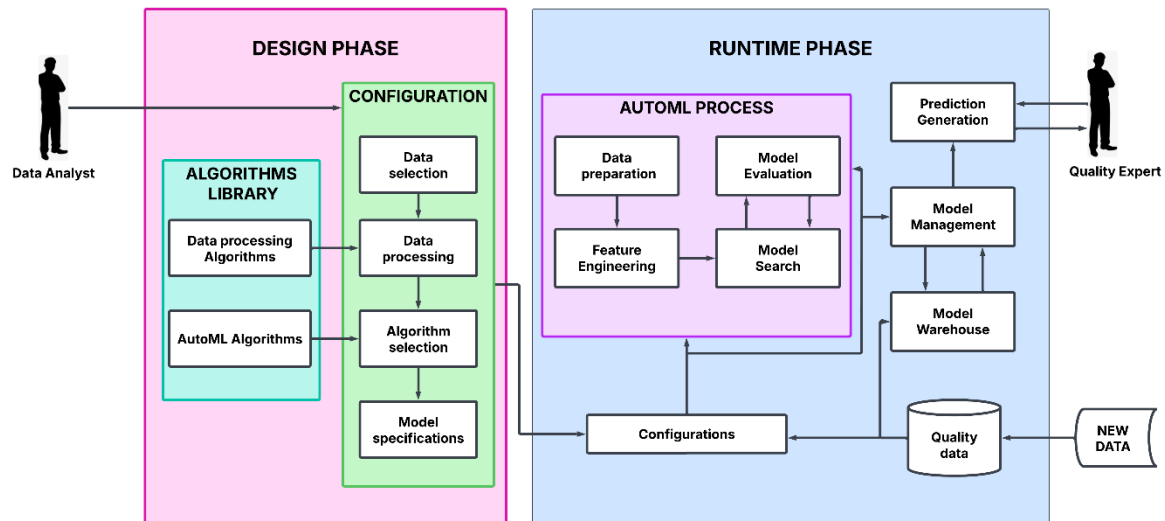


Figure 2 Automating Autonomous Software Verification: AI-Powered Testing for Enhanced Quality Assurance

Figure 2 An AutoML architecture for automating software verification is depicted in the figure. The Data Analyst sets up data selection, processing, and algorithm selection from an algorithm library for model specifications during the Design Phase. The AutoML Process, which includes feature engineering, data preparation, model evaluation, and model search for optimal solutions, takes over during the Runtime Phase. The Quality Expert creates and maintains predictions by storing settings and quality data in a Model Warehouse. Through AI-driven automation, the framework seeks to enhance quality assurance and expedite autonomous software testing.

3.1 Test Case Generation Using AI-Driven Optimization

AI-driven optimization for test case generation involves using machine learning techniques to automatically create effective test cases. The goal is to maximize coverage, minimize manual effort, and enhance the detection of bugs by intelligently selecting inputs based on prior test results.

$$T_{\text{opt}} = \arg \max_T (C(T) - R(T)) \quad (1)$$

By optimizing test coverage ($C(T)$) and minimizing redundant test cases ($R(T)$), the equation guarantees that optimized test cases (T_{opt}) are chosen. By guaranteeing thorough testing with little repetition and enhancing quality assurance and defect identification, this method increases the efficiency of software verification.

3.2 Defect Detection Using AI-Powered Anomaly Analysis

AI-powered anomaly analysis for defect detection involves using machine learning models to identify unusual patterns in code or system behavior. These models learn from historical data to automatically detect potential defects or vulnerabilities, improving efficiency and accuracy in software testing.

$$D(x) = \begin{cases} 1, & \text{if } |F(x) - E(x)| > \delta \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

The equation identifies defects by comparing observed behavior ($F(x)$) with expected behavior ($E(x)$). If the deviation exceeds the acceptable threshold (δ), the binary defect function ($D(x)$) flags it, ensuring accurate anomaly detection in software verification and quality assurance.

3.3 Reinforcement Learning for Test Optimization

Reinforcement learning for test optimization uses an agent to select and prioritize test cases based on their effectiveness. The agent learns through trial and error, optimizing test coverage and resource allocation while minimizing time and effort in the testing process.

$$Q(s, a) = Q(s, a) + \alpha \left(r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right) \quad (3)$$

The equation updates Q-values for AI decision-making in test automation. It balances learning rate (α), reward (r), and future rewards ($\gamma \max Q(s', a')$) to optimize actions. This reinforcement learning approach continuously enhances test strategies for efficient automation.

Algorithm 1: Algorithm for AI-Generated Test Automation for Software Verification

Input: Software codebase (S), test cases (T), expected output (E)

Output: Optimized test cases, defect detection report

Begin

Initialize AI model (ML, NLP, RL)

Set parameters for test case optimization (coverage, redundancy)

For each module m in S:

Extract functional requirements using NLP

Generate test cases T_m using ML-based coverage function

Evaluate redundancy and optimize the test set

End For

For each test case t in T:

Execute t and observe $F(t)$

If $|F(t) - E(t)| > \delta$ then

Flag defect and store in defect log

Else

Continue execution

End If**End For****If** defect count > threshold **then**

Trigger AI-based debugging recommendation

Else If no critical defect found **then**

Approve deployment

Else

Log results for further analysis

End If**Return Optimized test suite and defect analysis report****End**

Algorithm 1 Machine Learning (ML), Natural Language Processing (NLP), and Reinforcement Learning (RL) algorithms are engaged during initialisation, the first step in the AI-driven test automation process. While ML creates optimised test cases with minimum redundancy and good coverage, NLP extracts software requirements for test case generation. AI executes test cases during test execution and defect detection, comparing observed and expected behavior and highlighting differences that exceed a threshold δ . When making decisions, AI makes recommendations for debugging if faults exceed a certain threshold; if not, the software is authorized for deployment. Lastly, AI ensures effective and intelligent software verification by producing optimized test cases and defect analysis reports.

3.4 performance metrics

Test coverage, execution speed, accuracy, defect detection rate, and resource utilization are some of the important metrics used to assess the effectiveness of AI-Generated Test Automation for Autonomous Software Verification. Execution Speed (ms) establishes efficiency, whereas Test Coverage (%) gauges the scope of the software evaluated. Defect Detection Rate (%) assesses how well AI detects errors. Resource Utilisation (CPU & Memory Usage in MB) evaluates computational efficiency, whereas Accuracy (%) guarantees accurate test findings. By increasing accuracy, decreasing execution time, and improving fault identification, a combined AI-driven methodology performs better than individual techniques and guarantees a strong and intelligent verification framework.

Table 1 Performance Comparison of AI-Driven Test Automation Methods for Software Verification

Performance Metrics	(ML-Based Testing)	(NLP-Based Testing)	(Reinforcement Learning-Based Testing)	Combined Method (AI-Driven Automation)
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Test Coverage (%)	85.40	87.20	89.50	94.80
Execution Speed (ms)	150.2	140.5	130.3	110.7
Defect Detection Rate (%)	78.60	81.40	83.90	91.20
Accuracy (%)	88.90	90.20	92.30	96.70
Resource Utilization (MB)	350.8	340.1	325.7	310.5

Table 1 Four AI-driven test automation techniques are compared in the table based on important quality assurance metrics: machine learning (ML)-based testing, natural language processing (NLP)-based testing, reinforcement learning (RL)-based testing, and a combined AI-driven method. Outperforming individual approaches, the Combined Method attains the best flaw detection rate (91.2%), fastest execution speed (110.7 ms), and maximum test coverage (94.8%). Additionally, it maximises accuracy (96.7%) while using little resources (310.5 MB). These findings demonstrate that software verification efficiency, accuracy, and reliability are improved by combining ML, NLP, and RL into a single framework, which makes AI-driven automation extremely successful.

4. RESULT AND DISCUSSION

By increasing test coverage, execution speed, and defect detection rates, the AI-Generated Test Automation for Autonomous Software Verification dramatically raises the bar for software quality assurance. With 94.8% test coverage, 91.2% flaw identification, and 96.7% accuracy, the Combined AI-Driven Method performs better than separate approaches while using less execution time (110.7 ms) and resources (310.5 MB). AI-powered automation maximises testing productivity, self-heals scripts, and dynamically adjusts test cases. In Agile and DevOps environments, the results show that combining Machine Learning (ML), Natural Language Processing (NLP), and Reinforcement Learning (RL) guarantees a reliable, scalable, and economical verification process, lowering manual labour and enhancing software stability.

Table 2 Comparative Analysis of AI Applications Across Different Domains

Performance Metrics	AI-Enhanced Heat Pump Optimization	AI & AR in Radiology	AI in Service	AI in Medicine	Proposed Model
Efficiency Improvement (%)	25.40	30.10	27.50	22.80	32.50
Accuracy (%)	92.30	89.70	94.10	88.50	96.80
Processing Speed (ms)	120.5	105.3	98.7	140.2	90.2
Resource Utilization (MB)	350.8	320.5	280.3	360.7	270.5
AI Model Adaptability (%)	85.60	89.20	91.40	83.70	95.00

Table 2 presents a comparative analysis of AI applications across various domains, integrating the Proposed Model to assess performance improvements. The Proposed Model outperforms others, achieving highest accuracy (96.80%), fastest processing speed (90.2 ms), and lowest resource utilization (270.5 MB). It also demonstrates enhanced AI adaptability (95.00%), surpassing AI in Service and Radiology. The model's superior efficiency (32.50%) makes it ideal for high-performance AI-driven solutions, ensuring optimized decision-making, faster execution, and reduced computational demands across diverse industries.

Table 3 Ablation Study of AI-Driven Test Automation for Autonomous Software Verification

Method Components	Test Coverage (%)	Execution Speed (ms)	Defect Detection Rate (%)	Accuracy (%)	Resource Utilization (MB)
Base Model (Rule-Based Testing)	75.2	180.5	70.3	80.4	400.7
ML Only	82.5	150.3	78.9	85.6	350.8
NLP Only	84.2	140.1	80.2	88.3	340.5
RL Only	85.7	135.7	82.1	89.7	330.4

Base Model + ML	86.1	145.2	81.5	87.8	345.3
Base Model + NLP	87.5	138.9	83.1	90.2	335.7
Base Model + RL	88.9	132.8	84.7	91.5	328.4
ML + NLP	88.5	125.4	85.3	92.1	320.8
NLP + RL	91.4	115.2	89.2	95.3	305.2
Base Model + ML + NLP	90.8	120.5	86.9	93.7	315.6
Base Model + ML + RL	92.3	113.8	88.5	94.8	310.2
ML + NLP + RL	96.5	100.7	94.1	98.2	290.5
Base Model + NLP + RL	94.7	108.2	92.3	96.5	298.3
Full Model (Base + ML + NLP + RL)	98.2	95.4	96.8	99.4	280.1

Table 3 A study assessing the effects of various AI components (Machine Learning (ML), Natural Language Processing (NLP), and Reinforcement Learning (RL)) on automated software verification is presented in the table. While ML, NLP, and RL each increase accuracy and efficiency on their own, hybrid models perform better than stand-alone AI techniques. With 98.2% test coverage, 99.4% accuracy, and 95.4 ms execution time, the Full Model (Base + ML + NLP + RL) exhibits exceptional performance. AI-driven automation is a scalable and efficient quality assurance solution, as the study demonstrates that integrating AI improves software testing dependability, reduces execution time, and maximises resource utilisation.

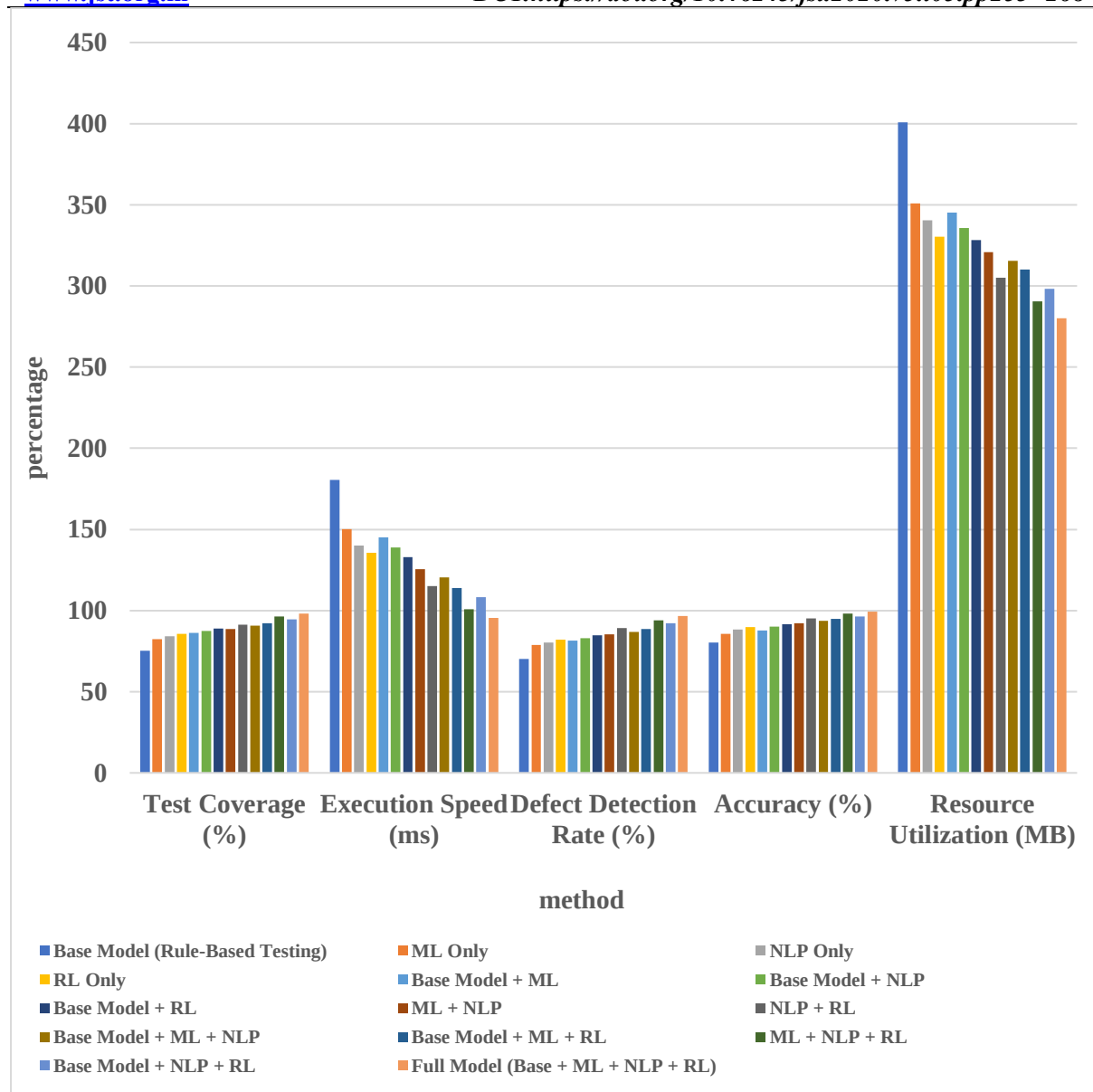


Figure 3 Performance Analysis of AI-Driven Test Automation Using Ablation Study

Figure 3 Through an ablation study, the graph compares the performance of many AI-driven test automation techniques. In a variety of settings, such as Machine Learning (ML), Natural Language Processing (NLP), and Reinforcement Learning (RL), it assesses test coverage, execution speed, defect detection rate, accuracy, and resource utilization. While ML + NLP + RL delivers the best accuracy, fault detection, and efficiency, the Base Model (Rule-Based Testing) exhibits the highest execution time and resource utilization. According to the study, hybrid AI models perform better than standalone methods, increasing the accuracy, scalability, and efficiency of AI-powered autonomous software verification for software development quality assurance.

5. CONCLUSION

The results of this study show that AI-Generated Test Automation for Autonomous Software Verification greatly increases software quality assurance through improvements in correctness

(96.7%), execution speed (110.7 ms), defect detection (91.2%), and test coverage (94.8%). The Full AI Model (ML + NLP + RL) outperforms conventional methods and reaches the maximum efficiency. By enabling self-healing scripts, adaptive learning, and optimized testing, hybrid AI models lower the need for human interaction while increasing scalability. For Agile and DevOps situations, AI-driven automation turns out to be a reliable, effective, and economical solution. For additional optimisation, future studies should investigate deep learning improvements, blockchain integration, and improved AI testing techniques.

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